

Module 07: Communication — Neural Resonance, Mirror Systems, and Social Cognition

Learning Objectives

1. Describe how the **mirror neuron system** supports action understanding and social inference through motor simulation.
2. Explain neural resonance and interpersonal synchrony as the neural basis of coupled Active Inference between agents.
3. Analyze the neural underpinnings of Theory of Mind, including the temporo-parietal junction and medial prefrontal cortex.

Introduction

Communication requires that isolated brains — each behind its own skull — achieve sufficient shared understanding to coordinate behavior. How does this happen neurally? This module examines three interacting systems: the **mirror neuron system** (which maps observed actions onto the observer's own motor repertoire), the **mentalizing network** (which represents other agents' beliefs and intentions), and the phenomenon of **interpersonal neural synchrony** (where interacting brains entrain to each other).

Key Concepts

1. Mirror Neurons and Motor Simulation

Mirror neurons, discovered by Rizzolatti et al. (1996) in premotor cortex of macaques, fire both when an animal performs an action and when it observes the same action performed by another. In Active Inference terms, mirror neurons implement **motor simulation** — the observer generates the same proprioceptive predictions as the actor, allowing the observer's generative model to "predict" the actor's intentions by simulating them internally.

In humans, the mirror neuron system (inferior frontal gyrus, inferior parietal lobule) provides the sensorimotor basis for understanding others' actions without explicit propositional reasoning.

2. Interpersonal Neural Synchrony

Hyperscanning studies (simultaneous recording of brain activity from multiple interacting individuals) reveal that interacting brains show **neural synchrony** — correlated oscillatory activity, particularly in the alpha and gamma bands. Hasson et al. (2012) demonstrated speaker-listener neural coupling: the listener's brain activity patterns converge with the speaker's, with a temporal lag corresponding to comprehension. Remarkably, successful communication correlated with tighter coupling, while miscomprehension correlated with decoupling.

In Active Inference terms, interpersonal synchrony is the neural signature of **generalized synchrony** — the two agents' generative models aligning through reciprocal prediction error exchange.

3. The Mentalizing Network

Theory of Mind — the capacity to attribute beliefs and intentions to others — depends on a distinct neural network: the **temporo-parietal junction (TPJ)**, **medial prefrontal cortex (mPFC)**, **posterior superior temporal sulcus (pSTS)**, and **temporal poles**. Saxe and Kanwisher (2003) showed that the TPJ is selectively activated when participants reason about others' beliefs, not simply about physical events.

In Active Inference terms, the mentalizing network implements the hierarchical component of the generative model that represents *other agents as inference systems* — modeling not just what another person does but what they believe, desire, and intend.

Clinical Connections

- **Autism and social communication:** Some accounts propose reduced spontaneous mirror activity or reduced mentalizing precision, leading to difficulties in automatic social inference (though compensatory explicit reasoning may develop).

- **Schizophrenia and paranoia:** Overactive mentalizing (higher-order ToM) may lead to aberrant attribution of intentions to others — e.g., perceiving neutral actions as threatening or conspiratorial.

Conclusion

Social communication is built on a neural infrastructure that implements coupled inference between brains — from the low-level motor resonance of the mirror system to the high-level belief modeling of the mentalizing network. Module 08 examines the temporal dimension of social and individual cognition — planning, decision-making, and the prefrontal cortex.